

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision R67 responds to the 10 October 2026 advisory report on R65. It retains the original NBO paper and adds operational prediction certificates, corrected complete-service evidence and a point-by-point response.

1. RESPONSE TO THE REFEREE

1.1. *Neural Bellman Operators — Revision R67*

We thank the referee for recognizing the state-lattice-free policy certificate and for identifying the additional evidence needed to assess the trained selector. We have revised the same Neural Bellman Operators paper. The title, own-future continuation construction, feasible witnesses, original Bellman benchmark and broader historical development are retained. The active article now connects a computable query decision to a certificate-aware training loss, and evaluates stronger controls, prospective reuse, actual continuous-law policy costs and the previously specified two-control extension.

The controlling report is the advisory review on branch review/econometrica-numerical-methods-r65-2026-10-10-34a5ce1, commit d548a97f461160296239ed6a973d4a91ef9d1fad. It reviews R65 commit 34a5ce17dc3ac4681b6d004eca00637f28104e28. We also reviewed the subsequently deposited R66 protocol and execution rather than treating R65 as the last repository work. R66 is not relabeled as an integrated paper or an external acceptance decision.

That review uncovered a numerical enclosure error in R66. The cumulative path-cost interval was initialized with aliased lower and upper arrays. In-place updates overwrote its running lower endpoint. An independent rational trajectory falls below a saved lower endpoint, so the issue is substantive even though it is small in magnitude. We retained the original records, repaired the accumulator, com-

mitted and froze the corrected sources, and reran the entire 212-process catalogue in R67. The fixed random indices are unchanged to isolate the correction; their repetition does not add independent observations. The active cost intervals and complete timings come from the corrected run, with the original R66 evidence and the counterexample preserved.

1.2. *B1, M1: The NBO contribution and the role of learning*

The revised article distinguishes the constructive own-future ReLU continuation from the trained query selector and from the method-neutral verifier. It does not infer neural computational superiority from an exact ReLU encoding. The original operator, construction, witness-preserving compiler and acquired-policy statements remain in the active theorem chain, with their arguments preserved in the supplement and the complete earlier editions.

The positive new contribution is operational. A strong-convexity upper chord and a global action-search lower bound can certify an actual proposed action before a new Bellman query. A margin loss bounds the endpoint work left unresolved by the proposal. A fixed-feature ReLU readout can minimize a convex certificate-aware objective with a directly computable optimization-gap certificate. These additions do not rely on successful nonconvex hidden-layer training or on the unknown location of the optimal action. They develop the existing NBO question rather than substituting a different paper or claiming exclusivity for generic safe improvement.

1.3. *B2, B3, M2: Strong conventional controls and complete cold-start work*

The source-frozen comparison contains ten selectors with the same original-law kernel, pre-query screen, acquired action rule and accuracy target. The fitted controls now include full quadratic, linear, additive piecewise-linear spline and nearest-action predictors. Unfitted controls include adaptive parabolic search, bisection, direct minimization of the upper chord, an analytic terminal-derivative proposal and a midpoint rule. The analytic derivative is only a proposal at earlier dates; it is not treated as their true optimal action.

Every fitted selector receives the same ninety-six action-label states for a given task and seed. Eight predetermined seeds and two clock repetitions produce the complete comparison catalogue. Training labels, fitting, imports, prediction, queries, path enclosures, output durability and process shutdown are all charged. The main comparison table reports complete process times rather than presenting an online query saving as a cold-start win. The supplement retains every fitting seed, optimizer warning and actual query count. Closely spaced clocks are not interpreted as statistically identified runtime differences.

The new upper-chord screen is offered to the conventional methods as well as the neural selector. It is therefore possible to assess prediction against a stronger analytic control, not merely against a deliberately weak initialization. Earlier conventional wins and the twenty worse pure ReLU policies remain in their original files and companion manuscripts. The current evidence is not used to rewrite those results.

1.4. B4, M3: A computable training-to-query connection

The earlier localization proposition contains an unknown action error and a derivative at the unknown optimum. It is retained as an analytical result, but the new section **Operational prediction certificates and readout training** supplies a different, computable condition.

For consecutive queried actions $l < u$, verified upper endpoints and a primitive strong-convexity bound define a quadratic upper chord C . If the outward value of C at the actual rounded proposal minus the existing global lower bound is at most the local tolerance, the proposal is certified without a new endpoint query. Its proof uses only the original objective's strong convexity and valid endpoints. A failed screen leaves the protected-refinement and full-policy guarantees intact.

The score is then used to define a margin loss. If a failed screen can require at most $K(z)$ further endpoint queries in context z , the additional count is bounded by $K(z)$ times the clipped squared margin loss. This is a deterministic statement for every specified context catalogue. A separate finite-family validation theorem gives an expected-work bound under an explicitly declared, independent, predictor-

independent context law. Recursively generated on-policy contexts are not assumed to satisfy that law merely because they are stored in a transcript.

The fixed-feature ReLU construction goes further. Nonnegative partition-of-unity features and a box readout produce a feasible action inside a fixed context interval. The squared positive-part certificate surplus, weighted by the fallback work cap, is convex in the readout. Its achieved objective bounds unresolved work, and its box dual gap bounds the optimization error. The deposited exact-rational implementation returns both quantities after finitely many conditional-gradient iterations. Exact tests include nonvacuous pre-query closure and successful certificate-aware fitting examples.

This constructive readout is distinct from unrestricted hidden-layer learning. The benchmark’s L-BFGS models are not relabeled as minimizers of the new objective. Their action-label mean squared errors, warnings and actual certificate progress remain separate empirical records. The theorem therefore supplies the requested operational connection without asserting an unexecuted population generalization result for the benchmark fits.

1.5. B5, M4: Prospective cached reuse, including failure handling

The new reuse catalogue fits once, writes an identified model and loads it for twenty-four predetermined new workloads. ReLU and quadratic seeds 6603 and 6607 were fixed in the protocol. Adaptive and bisection controls receive the same workload sequence. All model reads, digest checks, identity checks, predictions, verifier calls and durable outputs are charged.

At block index 7, the loaded acquisition-contract identifier is deliberately stale. The service rejects reuse and executes the conventional adaptive verifier. The stored model is not corrupted or silently repaired. The failed reuse and fallback are part of both the transcript and the complete measured process. This replaces a hypothetical reuse allowance by an executed failure path.

The supplementary table reports the first observed cumulative crossing against the matched adaptive repetition, a crossing that persists within the remaining observed window, and the final complete-process comparison. We distinguish a

1 prefix clock ending at a durable block from a process clock ending after shutdown. 1
 2 No prefix is called the complete cost of a stopping experiment that was not run. All 2
 3 declared seeds and repetitions are retained, including reversals and noncrossings; 3
 4 no amortization beyond the observed twenty-four blocks is extrapolated. 4

6 1.6. *B6, M5: Actual continuous-law costs of the callable controllers* 6

7 The corrected inference services evaluate the actual adaptive, bisection, ReLU 7
 8 and quadratic callable controllers at every date. They do not substitute an old 8
 9 tensor policy, a fixed-action rollout or a deterministic workload average. The fitted 9
 10 seed is fixed before evaluation. Initial states are uniform on the full cube and the 10
 11 innovations retain their original continuous laws. 11

12 Forty-eight-bit bins enclose complete continuous initial and shock coordinates. 12
 13 The acquisition rule is checked on the propagated state interval. Any ambiguous 13
 14 acquisition receives a full remaining-cost support enclosure rather than deletion 14
 15 or midpoint substitution. Marginal cost and paired difference intervals use out- 15
 16 ward endpoint samples, signed path supports and the stated empirical-Bernstein 16
 17 inequality. All four absolute costs and six differences at both tasks belong to one 17
 18 simultaneous family with error at most $1/100$. 18

19 There are 8,192 distinct common-path rows per task, hence 16,384 across the two 19
 20 tasks. Four method arms and repeated R66/R67 streams do not multiply this count. 20
 21 The random seed binds reproducibility; independent bin sampling is the explicitly 21
 22 stated statistical model. The repaired execution and independent path checks are 22
 23 prerequisites for using the intervals, not cosmetic postprocessing. 23

24 The current paired intervals remain unresolved in sign. The manuscript reports 24
 25 this finding rather than turning uniform accuracy or a tighter certificate into an iden- 25
 26 tified expected-cost improvement. The requested direct endpoint is now measured 26
 27 for the new policies, and its limitations are visible in the full contrast table. 27
 28 28

29 1.7. *B7, M6: Training variation and numerical variability* 29

30 30
 31 Eight fitting seeds replace the earlier two-seed design. Each has two separate 31
 32 complete-process timings; the latter are not counted as independent training ini- 32

tializations. The frozen catalogue retains all warnings, losses, model identities, query counts, screen outcomes and complete times. A warning does not authorize removing a run or selecting a different trained model.

The resulting evidence measures variation across the declared training seeds. It is not an optimizer-convergence probability for a population of initializations. The constructive readout proposition supplies a different guarantee: the achieved finite optimization certificate for the fixed features and contexts actually used. This distinction prevents numerical repetition from being used as a substitute for either a training theorem or an independent statistical sample.

1.8. *M8: Direct progress of routed and screened proposals*

The scientific kernel records old and new global lower endpoints, old and new upper witnesses, the actual queried action, the query interval and the routing indicator for every refinement. The audit checks that upper and lower progress sum to gap reduction and that the maintained bounds do not move in the wrong direction. It separately counts routed queries with no progress, upper-only progress, lower-only progress, simultaneous progress and successful screens without a new query.

The displayed progress table counts each distinct transcript once. Both original clock repetitions remain available. We do not infer distance to the true minimizing action from action-label error, nor do we claim a query was useful solely because it was routed. These diagnostics make the computational mechanism observable on the same original model and tolerance.

1.9. *B8, M7: More controls, innovations and horizon*

The manuscript gives a proved simplicial query for the already specified two-control, shared-capacity investment extension. A semiconcavity minorant is formed on each action triangle. Its lower bound is computed by an outward supporting plane, so an approximate quadratic minimizer has no certificate authority. Verified feasible upper witnesses and a continuous simplex cover give the same local-gap

premise as the scalar policy theorem. The proof includes two-innovation quadrature and finite rational recovery under the explicit precision slack.

The executed vector catalogue covers state dimensions two and eight and horizons two, three and four, with two controls and two independent continuous innovations. No state lattice or stored reference policy is loaded. Query counts, descendants, batch size, rational recovery and process memory are reported. The four designed initial states per service are identified as scaling paths, not as an estimate of population expected cost. The much larger horizon-four work is retained.

This is a multivariate-control extension of the verified NBO backend, not a trained two-output selector comparison. The work recursion includes the q^2 descendant factor. Batching is not mislabeled as depth-first memory or as elimination of the continuation tree.

1.10. B9, M9: Economic purchase rather than certificate ranking

The purchase account now has direct cost intervals for the new callable policies and actual complete receipts for both cold start and reuse. For a candidate's expected gain interval $[l, u]$, measured work difference ΔW , normalized value v , compute price p and adoption fee f , the net-benefit interval is $[vl - p\Delta W - f, vu - p\Delta W - f]$. Its positive lower endpoint licenses purchase; a negative upper endpoint rejects it; otherwise the choice remains unresolved.

A purchaser who requires only the stated uniform accuracy contract can instead select the least measured complete work among eligible catalogue services. That is a different decision from claiming an identified expected-cost gain. The paper keeps the two criteria distinct and reports normalized model quantities, not an invented empirical calibration of hardware, installation costs or welfare.

1.11. B10, M10: A focused article with the original development preserved

The active reading path is now the original NBO construction, primitive regularity, verified queries, operational prediction and training, the two-control extension and complete economic evidence. The main statements remain in the article. Longer inherited proofs are relocated verbatim to the technical supplement with explicit

cross-references. Prior source hashes and label locations are checked, and the preceding main article, supplement and four broader development documents accompany the revision as unchanged companions.

The title and subject remain Neural Bellman Operators. We have not replaced the paper by a generic framework, removed unfavorable comparisons or silently changed the economic law. The broader diffusion, recursive-preference and game results retain their original assumptions in the complete development. The present linear-expectation query theorem is not asserted to cover every nonlinear application.

1.12. *Verification and delivery*

The release checks the corrected source freeze, all 212 catalogue receipts, all saved path enclosures, the model and action identities, the prospective reuse sequence, and the cost-interval arithmetic. It re-queries the ninety distinct scalar comparison services from their saved models and checks their duplicate clock transcripts. Separately written rational primitives check fixed midpoint trajectories as additional diagnostics. The large vector continuation trees and all inference/reuse endpoint queries are not independently re-integrated by that publication replay.

Inherited and new exact tests, source-preservation checks, bibliography checks, compilation of the active and companion documents, and a clean archive rebuild are recorded in the delivery manifest. Visual inspection is recorded separately from compilation. These checks support reproducibility; they are not described as formal proof verification or an external journal acceptance decision.